Article

TRACE-MOEA: Constrained Power-Grid Portfolio Search with Preference-Guided Elitism and Run-Level Event Summaries

Yubin Lin, Jiyu Li, Xiaofei Ruan, Xiaoyu Huang, Dishan Yang

Economic and Technological Research Institute of State Grid Fujian Electric Power Co., Ltd., Fuzhou 350000, Fujian, China
* Correspondence: 18606932711@163.com (Y. Lin)

Abstract

Utilities reviewing large grid-project queues need constrained portfolio search and inspectable summaries of events generated during optimization. TRACE-MOEA combines preference-guided elitism with deterministic budget repair and quarantines event statistics from objectives and selection. On a reproducible five-objective, 120-project proxy benchmark, the complete framework reaches a descriptive pooled mean hypervolume of 0.17425 across seven scenarios and 30 seeds, 0.89% above NSGA-II. The per-scenario differences are most favorable in preference-emphasized settings, but removing preference adaptation changes pooled hypervolume by only 0.17% and its direct effect remains unresolved after cross-scenario correction. Matched-budget preference controls favor TRACE-MOEA over the implemented R-NSGA-II configuration, whereas matched one-output comparisons expose competing objective trade-offs. Normalization changes deterministic-rule ordering and therefore bounds cross-family claims. Released event fields report only run-level count and pool-position co-occurrence; the mean co-occurrence is 98.6%, but stable identifiers, event order, replacement flags, and replay state are absent. The results support an integrated budget-constrained proxy-search framework with measurable event summaries, not an independent preference-adaptation benefit, normalization-invariant decision superiority, or project-level lineage.

Keywords: power grid investment review; constrained portfolio search; multi-objective evolutionary algorithm; adaptive preference elitism; budget repair; event co-occurrence summary; hypervolume

1. Introduction

Grid operators face more candidate investments than available capital can support. The queue includes line reinforcement, protection and automation retrofits, storage, and renewable-support installations. Expansion-planning research offers mature tools for sizing and routing assets under uncertainty [1,2]. A review board faces a different task: select from predefined projects under a budget, balance reliability, renewable accommodation, and schedule risk, and justify each inclusion to regulators and stakeholders.

The instruments that boards actually use come from multi-criteria decision analysis (MCDA). The AHP and TOPSIS families dominate energy investment appraisal [3,4] and continue to be refined for grid asset prioritization at this journal [5]. Their strength is procedural transparency: weights, pairwise judgments, and closeness scores can all be minuted. Their structural weakness is independent project scoring, which overlooks budget crowding-out and portfolio-level trade-offs. Exact portfolio methods such as Robust

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Portfolio Modeling address this issue for small pools [6,7], but richer dependencies and larger combinatorial pools can make them computationally difficult. MOEAs offer scalable approximate search [8,9], although a converged front alone does not explain why a project appears in a recommendation.

TRACE-MOEA is designed to search for budget-constrained portfolios while instrumenting two review-specific operations. An adaptive preference-elitism layer maintains objective-weight vectors and can restore their best-response portfolios after environmental selection; every five generations, the vectors are perturbed and greedily reselected according to the dispersion of their induced responses. Deterministic repair removes low-benefit selections until the budget is met. During a run, the implementation generates repair-drop and preference-best-response payloads using pool-local integer positions. The released evidence reduces those payloads to per-run event counts and final-front candidate co-occurrence coverage, and no event statistic enters an objective, constraint, or selection decision.

Because genuine utility review records with feasibility labels are confidential, evaluation is conducted on a reproducible proxy benchmark. The pool of 120 projects is generated by published deterministic rules from RTS-GMLC zones, SimBench networks, and metadata from 40 cached public NERC reliability reports. Seven scenarios cover portfolio optimization, distribution-only review, reliability and renewable preferences, budget tightening, preference-aware support, and traceability. A five-objective formulation encodes the investment-effectiveness lens. A two-rung external-consistency ladder then checks the selected portfolios against a NERC-derived rule and against historical MISO MTEP16 project outcomes.

The paper makes three evidence-bounded contributions:

1. **A constrained portfolio-search architecture.** TRACE-MOEA combines a constrained non-dominated sorting kernel with adaptive preference elitism and deterministic budget repair. The independent contribution of preference elitism is evaluated separately through component ablation and matched preference-family controls.
2. **Run-level event summaries separated from optimization.** Each main-run row reports total event count and final-front candidate-position co-occurrence for in-memory repair-drop and preference-best-response events. These fields are quarantined from objectives and selection and measure event production and set overlap rather than lineage, replay, or human-review value.
3. **A reproducible evaluation that separates framework and mechanism claims.** Seven review scenarios, 30 seeded runs per stochastic method, component controls, a matched three-budget preference comparison, a one-output analysis, and normalization sensitivity distinguish global front coverage from preference-layer attribution and compromise selection. The complete framework is competitive under the reported proxy metric, while the direct preference-ablation effect remains unresolved and the public-record checks remain descriptive.

The evaluation therefore distinguishes framework-level performance, preference-layer attribution, and the informational scope of the event summaries throughout the paper.

Section 2 situates the work in three literatures. Section 3 defines the problem and the public benchmark. Section 4 details the algorithm. Section 5 describes the experimental protocol, Section 6 the results, Section 7 the discussion, Section 8 the limitations, and Section 9 concludes.

2. Related Work

Three research threads intersect in this work: investment portfolio decision methods for power grids, preference-guided evolutionary multi-objective optimization, and the traceability of algorithmic decisions.

2.1. Grid Investment Decisions: Optimization and Review Scoring

Transmission expansion planning treats the network topology itself as the decision variable and has been surveyed comprehensively from both the modeling [1] and the practice side [2]; stochastic engineering-economic formulations bring market and regulatory uncertainty into the objective [10]. The present work addresses a different sub-problem: selecting from a pre-specified list of candidate projects under a hard budget, which is the institutional form of most utility investment cycles.

For this sub-problem, the energy-domain literature relies heavily on MCDA. The surveys of Wang et al. [3] and Kumar et al. [4] document the use of AHP and TOPSIS across energy investment appraisal tasks, and this tradition remains active: recent studies combine fuzzy-AHP with distribution-transformer maintenance priorities [5] and explainable multi-criteria scoring with strategic energy-investment ranking [33]. Portfolio-explicit alternatives exist in operational research. Robust Portfolio Modeling (RPM) computes non-dominated project portfolios under incomplete preference information [6] and has been applied to infrastructure maintenance programming [7].

Recent power-system planning studies have strengthened the application layer through coordinated transmission-storage planning [29], interruption-cost estimation for reliability investment [30], grid-side storage investment-return analysis [31], and hybrid exact-heuristic treatment of non-analytical investment costs [32]. These studies provide increasingly detailed engineering or economic validation, but they optimize system expansion or asset configuration rather than a reviewable selection process for a large, pre-specified project queue. Hybrid grid-investment models also couple risk and benefit dimensions using MCDA fronts [11]. The present study addresses a narrower combination: portfolio-level search over more than one hundred candidates, adaptive preference elitism, active budget repair, and run-level event co-occurrence summaries reported independently of the optimization metric.

2.2. Preference-Guided Evolutionary Multi-Objective Optimization

Dominance-based MOEAs [8] and decomposition-based ones [9,12] treat all trade-off directions as equally interesting, which motivated two decades of research on preference injection into evolutionary multi-objective optimization. Representative approaches include reference-point guidance [13], achievement-scalarizing function methods [14], preference-modified dominance relations [15], and comprehensive taxonomies of a priori, interactive, and a posteriori preference articulation [16,17].

Two developments in this thread are particularly relevant to the present design. RVEA demonstrated that reference vectors can be adapted during the run to the geometry of the objective space [18], establishing run-time preference adaptation as a legitimate mechanism–though on continuous benchmarks with no constraint repair and no decision recording. The second is the field's empirical evaluation of preference guidance. Li et al. [19] showed that measuring preference-based EMO fairly is itself a hard problem, and a subsequent holistic study found that reference-point guidance can degrade search performance on some problem classes [20]. The present results extend that caution to a constrained binary portfolio problem: the adaptive preference-elitism layer contributes only weakly on top of a well-repaired constrained kernel. The study evaluates that mechanism together with deterministic budget repair and separately reported run-level event summaries; it does not claim that any one component is new in isolation.

2.3. Traceability and Explainability of Algorithmic Decisions

The explainable AI (XAI) literature has matured a vocabulary of transparency and post-hoc explanation [21] and an account of what human decision-makers accept as explanations [22], but the object of study is typically the learned predictor rather than the optimization pipeline. The energy-domain XAI survey of Machlev et al. [23] finds applications concentrated in forecasting and security assessment, not in optimization. For metaheuristics specifically, explainability has been identified as an open research direction rather than an available toolbox [24].

Provenance research supplies taxonomies of why, how, and where records are generated [25], while the W3C PROV recommendation standardizes derivation records [26]. The algorithmic-accountability literature further distinguishes process artifacts generated during computation from post-hoc reconstructions [27]. TRACE-MOEA generates repair-drop and preference-best-response payloads during each run and quarantines their aggregate reporting fields from the evaluation objective. The current release preserves counts and pool-local set-overlap coverage rather than a PROV-compliant, ordered, or replayable project-level record.

2.4. Differentiation from the Companion Study

The companion project `mintou_p6_bilonsga_project_review` (BiLo-NSGA [28]) uses the same public candidate generator and common benchmark, evaluation, and public-record backtest infrastructure but poses a different mechanism question. It modifies variation through insertion, substitution, and dependency-aware local moves under a hard budget. TRACE-MOEA studies selection-stage adaptive preference elitism, deterministic repair, and run-level event co-occurrence summaries in a five-objective review formulation. The projects share infrastructure and source records; their configurations, executions, run outputs, selected portfolios, statistical comparisons, and claims remain paper-specific.

The independent TRACE-MOEA question concerns four measured elements: constrained search over more than 100 grid candidates, adaptive preference elitism, explicit budget repair, and run-level event co-occurrence summaries quarantined from evaluation. It also tests whether selected portfolios align with public project outcomes rather than only with the benchmark objectives. Its empirical scope is the public proxy and two descriptive external-consistency checks; expert validation of real utility review remains future work.

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3. Problem Formulation and Public Benchmark

3.1. Investment-Effectiveness Review as Constrained Five-Objective Selection

Let there be n candidate projects. Candidate i carries a cost k_i, a reliability benefit r_i, a renewable-accommodation benefit g_i, a compliance score c_i (reflecting alignment with regulatory standards), and an evidence score e_i (quantifying how well the project is supported by documented reliability records). Its aggregate schedule-and-implementation risk and review quality are

$$\rho_i = 0.58\rho_i^{\text{sched}} + 0.42\rho_i^{\text{impl}},$$

and

$$q_i = 0.5c_i + 0.5e_i,$$

respectively. A review decision is a binary vector $x \in \{0,1\}^n$, with selected-set cardinality $N(x) = \sum_i x_i$.

The selected-project mean risk and mean quality use the implementation's explicit empty-portfolio conventions:

$$\bar{\rho}(x) = \begin{cases} \frac{\sum_{i=1}^{n} \rho_i x_i}{N(x)}, & N(x) > 0, \\ 1, & N(x) = 0, \end{cases}$$

and

$$\bar{q}(x) = \begin{cases} \frac{\sum_{i=1}^{n} q_i x_i}{N(x)}, & N(x) > 0, \\ 0, & N(x) = 0. \end{cases}$$

The investment-effectiveness review problem is formalized as minimization of

$$F(x) = \left(\sum_i k_i x_i, -\sum_i r_i x_i, -\sum_i g_i x_i, \bar{\rho}(x), -\bar{q}(x) \right),$$

subject to

$$\sum_{i=1}^{n} k_i x_i \leq B.$$

The normalized budget violation used by constraint domination is

$$v_B(x) = \max\left\{ 0, \frac{\sum_i k_i x_i - B}{B} \right\}.$$

The minus signs convert maximization objectives to minimization for the hypervolume computation. Cost and budget are expressed in the same synthetic cost units; v_B is a dimensionless budget-overrun fraction. Reliability, renewable accommodation, risk, compliance, and evidence are proxy indices, not physical or monetary units.

The fifth objective — mean quality — distinguishes investment-effectiveness review from a generic knapsack. It rewards portfolios containing well-evidenced, compliance-strong projects, a property that review boards must assess. Two standard notions are used throughout. A portfolio is non-dominated if no other portfolio is at least as good on all five objectives and strictly better on one. Hypervolume is the normalized objective-space volume dominated by a front relative to a fixed reference point and is the sole optimization-performance metric used here.

3.2. Candidate Pool from Public Data

Real review dossiers are not published, so the pool is generated by deterministic, released rules from three public sources. Table 1 summarizes the derivation; the complete source profile is included in the supplementary evidence package.

RTS-GMLC zone aggregates (load, branch ratings and outage rates, generator capacities and renewable shares) produce three archetypes per zone — transmission reinforcement, reliability automation, and renewable support — with costs and benefits as analytic functions of the aggregates. The sixteen highest-stress SimBench subnets each yield a feeder-reinforcement, a storage-flexibility, and a protection-automation candidate driven by subnet load, line statistics, and the distributed-resource gap. The metadata of 40 cached public NERC reliability documents (28 event reports) then adjusts attributes at the archetype level: event-report counts raise reliability values uniformly, inverter-based-resource and storage mentions raise the renewable attribute for the two renewable-facing

Table 1. Candidate pool derivation from the released public-source profile.

Public Source	Used Artifact	Candidates	Archetypes
RTS-GMLC	Zone aggregates of bus, branch, and generator source data	72	Transmission reinforcement; reliability automation; renewable support
SimBench	Complete mixed network, 16 highest-stress subnets	48	Feeder reinforcement; storage flexibility; protection automation
NERC / C2GES cache	Metadata of 40 public reliability documents	Attribute adjustment	—
Total		**120**	6 archetypes

Table 2. The seven review scenarios in the accepted machine-readable configuration. Stakeholder weights parameterize scalarizing baselines and seed one preference vector of TRACE-MOEA; they never enter the evaluation metric.

Experiment ID	Pool	Budget	Weight Emphasis
benchmark_portfolio_optimization	Full (120)	1.00 x B	Balanced
distribution_project_review	SimBench candidates only	1.00 x B	Balanced
reliability_driven_review	Reliability-related archetypes	1.00 x B	Reliability +0.22
renewable_accommodation_review	Renewable/storage archetypes	1.00 x B	Renewable +0.24
budget_ranking_stability	Full	0.88 x B	Balanced
preference_aware_support	Full	1.00 x B	Compliance +0.16
traceability_evaluation	Full	1.00 x B	Evidence +0.28, compliance +0.12

archetypes, and document counts feed the evidence score. The result is 120 candidates in six archetypes with zone- and feeder-level dependency-group structure.

The epistemic status of this pool must be stated explicitly: it is an engineering-plausible, fully reproducible proxy whose every attribute traces to public data through published rules. It contains no expert-labeled review outcomes, and its cost coefficients are synthetic units calibrated to grid-case branch and generation statistics rather than to actual utility expenditure records (see Section 8).

3.3. Review Scenarios

Seven experiments vary the candidate pool composition and the assumed stakeholder weighting while the evaluation rule remains fixed (Table 2). Bounds are frozen separately for each scenario from its method-independent reference set, and every scenario uses the normalized reference point $(1.1, \ldots, 1.1)$. The nominal budget is $B = 1160$ synthetic cost units across all scenarios except the budget-stability scenario, where it is tightened to $0.88B$.

3.4. Shared-Pipeline Declaration

The candidate-generation code of Section 3.2 is a shared artifact used by this paper and by the companion project `mintou_p6_bilonsga_project_review` (BiLo-NSGA [28]). The projects also share source corpora, common execution and evaluation utilities, and public-record backtest infrastructure. The five-objective p5 formulation, p5 scenario and method configurations, executions, run outputs, selected portfolios, statistical comparisons, and claims are paper-specific. The Data Availability statement records this boundary.

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4. TRACE-MOEA

Figure 1 separates optimization state from event-reporting state. Adaptive preference elitism and deterministic budget repair operate around a constrained NSGA-II kernel,

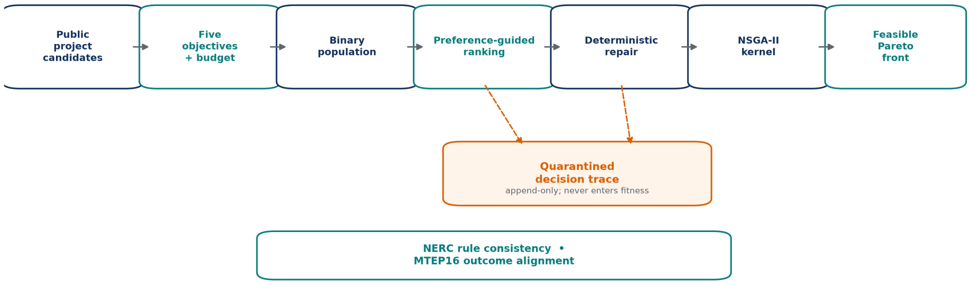

Figure 1. TRACE-MOEA architecture and quarantine invariant. Dashed one-way arrows write preference-best-response and repair-drop payloads to the in-memory event list; released run rows retain only count and pool-position co-occurrence summaries. The external checks assess returned portfolios but do not train or tune the optimizer.

while their event payloads are appended to an in-memory per-run list. The event channel has no return edge: no event count, coverage statistic, or event attribute enters an objective, constraint, fitness, or selection rule.

The algorithm augments a constrained non-dominated sorting kernel with two switchable search components, adaptive preference elitism and deterministic budget repair, and instruments both with a quarantined in-memory event list. Component ablations and the bare-kernel control separate their optimization and event-summary behavior in Section 6.2.

Implementation-contract notation. A portfolio is $x \in \{0,1\}^n$, with minimized five-objective vector, total cost, and dimensionless normalized violation

$$F(x) = [f_1(x), \ldots, f_5(x)]^\top, \qquad C(x) = \sum_{j=1}^{n} k_j x_j, \qquad v_B(x) = \max\left\{0, \frac{C(x) - B}{B}\right\}.$$

Thus k_j, $C(x)$, and B share synthetic cost units; v_B has no units. Constraint dominance is

$$x \prec_c y \iff [v_B(x) = 0 < v_B(y)] \vee [v_B(x) = v_B(y) = 0 \wedge x \prec_P y] \vee [v_B(x), v_B(y) > 0 \wedge v_B(x) < v_B$$

Preference scoring uses a *generation-local* normalization distinct from the fixed evaluation normalization in Section 5.2. For combined parent–offspring pool R_t,

$$\hat{f}_{tq}(x) = \frac{f_q(x) - \min_{y \in R_t} f_q(y)}{\max\{\max_{y \in R_t} f_q(y) - \min_{y \in R_t} f_q(y), 10^{-9}\}},$$

and, for simplex weight w_k,

$$\psi_{tk}(x) = \sum_{q=1}^{5} w_{kq} \hat{f}_{tq}(x) + \lambda_{\text{pref}} v_B(x), \qquad \lambda_{\text{pref}} = 10, \qquad x^\star_{tk} = \text{firstargmin}_{x \in R_t} \psi_{tk}(x).$$

The penalty is dimensionless because both terms are dimensionless. In the full method, initialization and offspring are repaired before R_t is formed, so $v_B = 0$ and this penalty does not alter its best-response ordering; it remains active in repair-disabled configurations. For scenario weights α, the first preference vector is

$$w_1 = \frac{(\alpha_{\text{cost}}, \alpha_{\text{rel}}, \alpha_{\text{ren}}, \alpha_{\text{risk}}, \alpha_{\text{comp}} + \alpha_{\text{evid}})}{\alpha_{\text{cost}} + \alpha_{\text{rel}} + \alpha_{\text{ren}} + \alpha_{\text{risk}} + \alpha_{\text{comp}} + \alpha_{\text{evid}}}.$$

The scenario's separate load-support weight is not a p5 objective and is not mapped into w_1. Seven other weights are sampled independently from Dirichlet$(1, \ldots, 1)$. Every fifth generation, each current weight is perturbed by

$$w'_k = \frac{|w_k + \varepsilon_k|}{\|w_k + \varepsilon_k\|_1}, \qquad \varepsilon_k \sim \mathcal{N}(0, 0.1^2 I),$$

where the absolute value is componentwise; this is the executed absolute-value-plus-renormalization rule, not Euclidean projection. From the $2K$ original and perturbed candidates, one weight is seeded uniformly at random and the remaining weights are added greedily by maximum distance from the already retained response set. This is a randomized greedy max–min heuristic, not a global dispersion optimum.

The five-objective mapping above applies directly to TRACE-MOEA and controls that retain all five search objectives. In `NoReliabilityFeatures`, `NoRenewableFeatures`, and `NoScheduleRisk`, the preserved implementation removes one objective coordinate but initializes the first scenario-derived preference vector by truncating its first four entries rather than remapping weights to the surviving objective indices; the other seven vectors are sampled directly in four dimensions. Each named objective-hiding condition therefore changes both objective visibility and the first-vector parameterization. These three conditions are combined-configuration stress tests, not isolated objective ablations.

Budget repair uses the code-level load-support attribute ℓ_j and the quality index q_j:

$$s_j = \frac{r_j + g_j + \ell_j + q_j}{\max\{k_j, 1\}}, \qquad j^- = \text{firstargmin}_{j:x_j=1}\, s_j,$$

and repeats $x_{j^-} \leftarrow 0$ until $v_B(x) = 0$. Event append is one-way: if A_h is the in-memory event list and e_h a generated payload, then $A_{h+1} = A_h \,\|\, e_h$, while no A_h field is passed to objective, violation, preference-score, repair-score, or environmental-selection functions. This is a data-flow invariant, not a probabilistic independence claim.

Finally, to avoid overloading λ_{pref}, five-dimensional Lebesgue measure is denoted by μ_5. Standard hypervolume is

$$HV(\mathcal{P}; r) = \mu_5\left(\bigcup_{x \in \mathcal{P}} \prod_{q=1}^{5} [\tilde{f}_q(x), r_q]\right),$$

with method-independent evaluation bounds and reference point fixed before comparisons.

4.1. Constrained Non-Dominated Sorting Kernel

The custom kernel keeps exactly 40 population rows for 40 update generations. For each initial row, a density $p_i \sim U(0.03, 0.15)$ is drawn and each candidate bit is Bernoulli(p_i). Each child draws two parent-row indices independently with replacement; a Bernoulli(0.5) mask chooses the parent at every coordinate, and bit-flip mutation is then applied independently at rate $1/n$. There is no separate 0.9 crossover gate in the custom implementation.

Parents and repaired children form an 80-row union. Constraint-dominated non-dominated sorting admits complete fronts, and descending crowding distance truncates the last admitted front. No authored secondary criterion resolves exact crowding ties; the executed NumPy ordering is therefore part of the implementation rather than a scientific preference. Environmental selection produces 40 distinct *union row indices*. Preference replacement preserves the 40-row invariant. Binary genotypes are not deduplicated during search, so equal portfolios may occupy different rows. Only the returned feasible population is deduplicated lexicographically before its non-dominated front, hypervolume, and coverage are computed.

4.2. Adaptive Preference Elitism

The layer maintains $K = 8$ objective-weight vectors $w_1, \ldots, w_K$ on the five-dimensional simplex, initialized and normalized as specified above. It acts twice per generation cycle:

- **Preference elitism.** For each w_k, `firstargmin` selects a best-response row from the parent–offspring union. If that row index is absent after environmental selection, a seeded uniformly random selected slot is replaced by it. Later weights can evict a row restored for an earlier weight. No crowding-, worst-score-, or age-based eviction rule is used, and the payload does not record whether replacement occurred or which row was evicted. Independently of replacement, the implementation appends a `preference_elite` payload with generation, weight-vector index, and the pool-local integer positions selected in the best response. The resulting $8 \times 40 = 320$ records per full run are best-response records, not injection or eviction counts.
- **Adaptive re-selection.** Every five generations, the K current and K absolute-value-perturbed weights are evaluated on the same generation-local normalized union. A seeded uniformly random candidate starts the retained set; greedy max–min response dispersion then adds weights until K remain. Exact dispersion ties use the first candidate returned by `argmax`. The procedure is stochastic and greedy, and neither the initial candidate nor the retained-vector indices are exported.

The layer is designed for review settings in which a decision maker cannot specify exact reference points in advance, an assumption common in preference-guided EMO (Section 2.2). It identifies weighting directions that yield distinct feasible portfolios and can preserve representatives from those directions. Whether this mechanism improves hypervolume is evaluated separately; Section 6.2 shows that its isolated pooled effect is small and unresolved after cross-scenario correction. This rationale is an algorithmic design hypothesis, not a validated theory of human preference elicitation or reviewer behavior; no interaction data were collected.

4.3. Budget Repair

Any initial or offspring portfolio that exceeds the budget is repaired by iteratively dropping the selected candidate with the smallest s_j: raw reliability, renewable, load-support, and quality proxies are summed and divided by `max(cost, 1)`. These inputs are not normalized, so the rule is an explicitly disclosed proxy heuristic rather than a unit-invariant benefit–cost estimator. Selected indices are enumerated in ascending pool order and `argmin` takes the first minimum; exact score ties therefore drop the smallest pool-local index. Each executed drop appends an in-memory event with the generation and that local index. Conditional on a portfolio and pool ordering, the drop sequence is deterministic.

The purpose of explicit repair is to keep the population inside the fundable region where the review question is actually posed, rather than leaving constraint pressure to selection alone. The ablation `Ablation-NoFeasibilityRepair` disables this operator and relies entirely on constraint-dominated sorting for budget feasibility.

4.4. Implemented Run-Level Event Records and Co-Occurrence Counters

Within each run, the implementation appends two event-record types to an in-memory list: `repair_drop` contains (`gen`, `event`, `item`), and `preference_elite` contains (`gen`, `event`, `pref`, `items`). The `item` and `items` values are zero-based positions in the current experiment pool. Although the candidate object has a stable `cid` field, that field is not written to these records or the released run table. Consequently, the current evidence cannot link an event position across differently filtered pools or establish lineage.

- **Defined scope.** Every executed repair drop produces one event record, and every weight vector produces one preference-best-response record per generation. Crossover, mutation, actual replacement, eviction, parentage, and retained-weight updates are not logged. A `preference_elite` record therefore identifies a scored best response but does not establish a population change.
- **Quarantine.** No event statistic (count, coverage, or payload attribute) appears in any objective function, constraint inequality, or selection criterion. The event fields are descriptive and are reported separately in Section 6.3.
- **Counters.** Let D be the number of executed repair drops, U_A the set of pool-local positions appearing in any generated payload, and $U_{\mathcal{P}}$ the set of positions selected in at least one deduplicated final feasible non-dominated portfolio. The released fields are

$$\texttt{trace\_event\_count} = |A| = D + KT = D + 320, \qquad \texttt{decision\_coverage} = \frac{|U_A \cap U_{\mathcal{P}}|}{\max\{1, |U_{\mathcal{P}}|\}},$$

for the full method with $K = 8$ and $T = 40$. The run field `local_move_count` equals D for p5; there is no released per-type event counter or replacement counter. Across 210 proposed-method runs, the means are 1126.25 total records, 806.25 repair drops, and 0.985688 coverage. Coverage is candidate-position co-occurrence after aggregating over the whole run, not same-portfolio causation, chronology, lineage, replay completeness, or human utility.

- **Released summary only.** The evidence package does not persist the ordered payload list. It retains only the two run-level fields above, so readers cannot reconstruct event order, map events to stable project identifiers, identify an evicted portfolio, or replay the population history from the release.

4.5. Algorithm Outline

The complete algorithm is summarized as:

```
TRACE-MOEA(pool, B, seed):
  for each of 40 rows: draw p ~ U(0.03, 0.15), sample Bernoulli(p) bits
  repair each row to B; append one repair_drop(gen=0, item=local_index) per drop
  preferences = mapped scenario vector + 7 Dirichlet(1) vectors
  for gen = 1 .. 40:
    draw two parent rows with replacement for each child
    choose each bit by a Bernoulli(0.5) parent mask; flip each bit with prob. 1/n
    repair every child to B; append repair_drop records
    R = parents followed by children
    selected = constraint-dominated NDS; crowding-truncate to 40 row indices
    normalize objectives by current R minima and ranges
    for k = 0 .. 7:
      best = first row minimizing w[k] * normalized_objectives + 10 * violation
      if best row index is absent from selected:
        replace one seeded-uniformly-random selected slot with best
      append preference_elite(gen, pref=k, items=best's pool-local positions)
    if gen mod 5 == 0:
```

```
      form absolute-value Gaussian perturbations and L1-renormalize
      seed one candidate weight at random; greedily retain 8 by response dispersio
    population = R[selected]                                     // always 40 row
  deduplicate feasible returned rows; retain their non-dominated front
  write only total event count and final-front position co-occurrence coverage
```

4.6. Configuration and Reproducibility Boundary

The generated main-run configuration records the seven experiment names, method names, 30 seeds, population size 40, generation limit 40, and the fixed hypervolume protocol. It does not serialize $K = 8$, $\lambda_{\mathrm{pref}} = 10$, the $U(0.03, 0.15)$ initialization density, mutation rate $1/n$, five-generation update cadence, perturbation scale 0.1, random eviction, tie behavior, or event-record fields. Those executed values are constants in the preserved implementation and are disclosed above; the configuration alone is not a replay contract. Legacy configuration terminology refers to the weight-update mechanism and an ephemeral in-memory event list; it does not establish a separate coevolving solution population, a released ordered record, lineage, or replay. The released run rows also omit objective-evaluation counts and actual preference-replacement counts. The matched-output and sensitivity stage has its own accepted machine-readable configuration and independent same-seed reproduction; it does not restore state missing from the original runs.

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5. Experimental Setup

5.1. Methods

Table 3 lists the sixteen methods: the proposed algorithm, seven baselines covering evolutionary, preference-based, and MCDM families relevant to grid review, and eight ablations designed to isolate individual components and objective-visibility effects.

Evolutionary baselines are pymoo implementations on the same binary objective problem and use the same row-wise $U(0.03, 0.15)$ initialization-density sampler. NSGA-II and R-NSGA-II are explicitly configured with population 40 and `n_gen=40`; they use Boolean two-point crossover and bit-flip mutation, but their operator probabilities are not overridden. The available run configuration does not pin the pymoo version, so those library-default probabilities cannot be certified as archival constants. MOEA/D instead uses the 35 Das–Dennis directions generated for five objectives and three partitions, 10 neighbors, neighbor-mating probability 0.7, and a $10^4 v_B$ objective penalty because this call exposes no inequality constraint. The archive retains no `n_eval` field, so equal generation labels do not establish identical objective-call budgets.

R-NSGA-II receives one scenario-derived raw objective-space reference point. With

$$a = (\alpha_{\mathrm{cost}}, \alpha_{\mathrm{rel}}, \alpha_{\mathrm{ren}}, \alpha_{\mathrm{risk}}, \tfrac{1}{2}(\alpha_{\mathrm{comp}} + \alpha_{\mathrm{evid}})), \quad \eta = \frac{a}{\max_q a_q},$$

the mapping is $z^{\mathrm{pref}} = 0.75\mathbf{1} - 0.55\eta$ in the fixed normalized coordinates of Section 5.2 and $p^{\mathrm{ref}} = l + z^{\mathrm{pref}} \odot (u - l)$ in raw objective units. The code passes p^{ref}, `epsilon=0.01`, and population 40 to R-NSGA-II. It does not pass fixed ideal/nadir arrays or override the library's normalization and distance weights. Per-generation internal ideal/nadir values and the executed pymoo version are not serialized, so the available evidence cannot reconstruct them; the fixed (l, u) bounds govern reference-point construction, the reported distance, and hypervolume, but should not be described as R-NSGA-II's internal survival bounds. Scalarizing baselines consume the scenario stakeholder weights of Table 2. No baseline receives a pool restriction or cardinality cap.

Table 3. Method matrix from the accepted machine-readable configuration.

Method	Role	One-line Description
TRACE-MOEA	Proposed	Constrained kernel + adaptive preference elitism + budget repair + run-level event co-occurrence summaries
NSGA-II	Baseline	pymoo, constrained, binary encoding
R-NSGA-II	Baseline	reference-point NSGA-II using the declared scenario preference point
MOEA/D	Baseline	pymoo, Tchebycheff decomposition, budget violation as penalty
AHP-TOPSIS	Baseline	Consistent AHP matrix, TOPSIS closeness ranking, greedy budget fill
Weighted Sum	Baseline	Weighted-score ranking, greedy budget fill
Greedy BCR	Baseline	Benefit-cost ratio greedy fill under budget
Random Feasible	Baseline	Random-permutation greedy fill
Ablation-NoFeasibilityRepair	Ablation	Repair disabled; constraint domination only
Ablation-NoPreferenceRanking	Ablation	Preference-adaptive layer disabled
Ablation-NoReliabilityFeatures	Combined control	Reliability objective hidden; preserved first-vector truncation also changes the surviving weight mapping
Ablation-NoRenewableFeatures	Combined control	Renewable objective hidden; preserved first-vector truncation also changes the surviving weight mapping
Ablation-NoScheduleRisk	Combined control	Risk objective hidden; preserved first-vector truncation also changes the surviving weight mapping
Ablation-SingleObjective	Ablation	Search on scalarized weighted sum
Ablation-NSGA2Only	Combined control	Bare kernel: no repair, no preference layer, no event records
Ablation-SmallProjectPool	Ablation	Candidate pool reduced to one third (~40 candidates)

Three combined controls (`NoReliabilityFeatures`, `NoRenewableFeatures`, and `NoScheduleRi` hide one objective from the search while evaluation remains the full five-objective hypervolume. Because the preserved implementation also truncates rather than remaps the first scenario-derived preference vector, their outcomes identify the joint effect of objective hiding and that parameterization change; they do not isolate the importance of the named objective. Three ablations remove algorithmic components (`NoFeasibilityRepair`, `NoPreferenceRanking`, and the combined `NSGA2Only` control), and two stress the problem structure (`SingleObjective` and `SmallProjectPool`).

5.2. Protocol, Metric, and Statistics

The archive contains 16 methods x 7 scenarios x 30 invocations, or 3360 rows. Twelve opponents are stochastic: four baselines (NSGA-II, R-NSGA-II, MOEA/D, and Random Feasible) and eight ablations. AHP-TOPSIS, Weighted Sum, and Greedy BCR are deterministic rules. Their 30 identical invocations are retained for a rectangular provenance table but provide an effective sample size of one per scenario. The candidate pool, budget, and scenario weights are fixed within each scenario.

The metric is the standard hypervolume of the feasible non-dominated front. Objective values are normalized by fixed per-scenario bounds computed once from a seeded, method-independent reference set consisting of the empty portfolio, all singleton portfolios, and 2048 random feasible portfolios. Singleton rows are not feasibility-filtered, so a singleton whose synthetic cost exceeds the budget can widen the empirical bounds even though it cannot enter a feasible returned front. If m_j and M_j are the reference-set minimum and maximum, the executed bounds are padded by 5%: $l_j = m_j - 0.05(M_j - m_j)$ and $u_j = M_j + 0.05(M_j - m_j)$. These are empirical reference bounds, not theoretical ideal and nadir values. The reference point is 1.1 in each normalized dimension. An experiment that returns no feasible portfolio scores zero. The three deterministic ranking rules produce one portfolio per scenario and appear as single markers in Figure 2. Random Feasible is stochastic and is analyzed by seed.

Seed-level inference uses two-sided Mann–Whitney U tests comparing TRACE-MOEA with each of the twelve stochastic opponents per scenario ($n = 30$ per group). P-values are Holm-corrected within that 12-comparison scenario family, with significance at $\alpha = 0.05$. We report rank-biserial effect sizes and 5000-resample bootstrap confidence intervals for mean differences. The intervals are pointwise and are not multiplicity-adjusted. Comparisons with the three deterministic rules are descriptive point gaps and are not assigned inferential p-values. For each named ablation family discussed across all seven scenarios, the raw scenario p-values receive a second Holm correction over those seven contrasts. This correction does not turn a combined configuration into an isolated component estimand. The pooled means across heterogeneous scenarios are descriptive and are not assigned a pooled hypothesis test.

To remove implementation ambiguity, objective j is normalized with method-independent bounds (l_j, u_j) as

$$\tilde{f}_j(x) = \min\left\{1, \max\left[0, \frac{f_j(x) - l_j}{u_j - l_j}\right]\right\}.$$

For a feasible non-dominated front $\mathcal{P}$ and reference point $z^{\mathrm{ref}} = (1.1, \ldots, 1.1)$, the reported quality score is

$$HV(\mathcal{P}) = \mu_5\left(\bigcup_{x \in \mathcal{P}} [\tilde{F}(x), z^{\mathrm{ref}}]\right).$$

For two seed samples with ranks R_i, the first-sample statistic is

$$U_1 = \sum_{i=1}^{n_1} R_i - \frac{n_1(n_1+1)}{2},$$

and ordered raw p-values are adjusted monotonically by

$$p_{(i)}^{\mathrm{Holm}} = \max_{k \le i} \min\left\{1, (M - k + 1)p_{(k)}\right\}.$$

The evaluation depends only on candidate attributes and fixed normalization bounds; preference vectors, event-record variables, and method-owned parameters do not enter the hypervolume metric. This separation is necessary because the implemented event record is an output to be inspected, not a reward to be optimized.

The matched-output extension consumes the compromise already selected in every preserved main-run row by the shared minimum-normalized-objective-sum rule. It retains 30 seeded rows for each of TRACE-MOEA, NSGA-II, R-NSGA-II, and MOEA/D, but collapses each deterministic rule to one unique method–scenario output, yielding 861 analysis rows. Because the quality coordinate of the selected compromise was not serialized, the matched analysis reports the available cost index, reliability, renewable support, risk, and portfolio size rather than reconstructing a single-point hypervolume. Full-front hypervolume is retained only as context.

The bound/reference rerun covers TRACE-MOEA, NoPreferenceRanking, and the three deterministic rules: $2 \times 7 \times 30 + 3 \times 7 = 441$ unique runs. Besides the reported clipped score, it computes the same empirical normalization without clipping, bounds expanded by 25% of their padded span on both sides, conservative definition-derived bounds, and reference points 1.1 and 1.2. For cost c, additive reliability r, additive renewable support g, portfolio risk ρ, quality q, and budget B, the conservative bounds are

$$l^A = (0, -\textstyle\sum_i r_i, -\sum_i g_i, \min\{1, \min_i \rho_i\}, -\max_i q_i), \qquad u^A = (B, 0, 0, \max\{1, \max_i \rho_i\}, 0).$$

Table 4. Descriptive pooled leaderboard over seven scenarios. Stochastic methods have 210 runs; deterministic-rule rows summarize seven unique outputs retained as repeated provenance rows. Event-record count and position-co-occurrence columns are run-level descriptive summaries only.

Method	Role	Mean HV	Std HV	Mean Runtime (s)	Event Records/Run	Position Co-occurrence
TRACE-MOEA	**Proposed**	**0.17425**	**0.00635**	**0.113**	**1126**	**0.986**
Ablation-NoScheduleRisk	Combined control	0.17403	0.00702	0.107	1125	0.986
Ablation-NoPreferenceRanking	Ablation	0.17396	0.00564	0.109	803	0.966
Ablation-NoFeasibilityRepair	Ablation	0.17300	0.00652	0.106	320	0.865
NSGA-II	Baseline	0.17270	0.00642	0.146	—	—
Ablation-NSGA2Only	Combined control	0.17249	0.00652	0.102	—	—
Ablation-NoRenewableFeatures	Combined control	0.16930	0.00375	0.106	919	0.964
AHP-TOPSIS	Baseline	0.13468	0.00269	0.065	—	—
R-NSGA-II	Baseline	0.11980	0.02671	0.192	—	—
Ablation-SingleObjective	Ablation	0.11617	0.02637	0.112	1611	0.984
Ablation-NoReliabilityFeatures	Combined control	0.09081	0.02124	0.106	915	0.977
Random Feasible	Baseline	0.08064	0.02261	0.065	—	—
Ablation-SmallProjectPool	Ablation	0.06920	0.01280	0.079	654	0.970
Greedy BCR	Baseline	0.05576	0.03248	0.065	—	—
Weighted Sum	Baseline	0.04105	0.01991	0.065	—	—
MOEA/D	Baseline	0.01954	0.01379	0.375	—	—

Clipping incidence uses a 10^{-12} numerical tolerance; the worst strict analytic-bound underflow before applying that tolerance was -3.36×10^{-17}. All 441 reported-HV cells reproduce the preserved values at eight decimals, and a second output directory matches all 17 path-independent artifacts byte-for-byte.

Finally, a one-factor-at-a-time sensitivity scan uses 30 common seeds per cell. It varies risk aggregation (portfolio mean versus maximum), the compliance share in the quality objective (0.25, 0.50, 0.75), preference-vector count ($K = 4, 8, 16$), and the seeded preference profile (balanced, reliability, renewable, traceability). All nine TRACE cells are reported; NoPreferenceRanking is rerun for the registered and three formulation cells, yielding 390 runs. The primary sensitivity readout is analytic-bound full-front hypervolume at reference point 1.2, with a matched single-point readout secondary. No p-values are computed. The public-record backtests are not rerun and retain their descriptive scope because no portfolio-level randomization family is introduced. New pymoo 0.6.2 front reruns could not be executed under one compatible `moocore`/CFFI host ABI, so the stage does not substitute another pymoo version or extend bound sensitivity to those preserved comparator fronts.

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6. Results

6.1. Main Comparison

TRACE-MOEA reaches mean feasible-front hypervolume 0.17425 (standard deviation 0.00635), 0.89% above NSGA-II at 0.17270. This is a descriptive scenario-pooled framework comparison rather than a pooled hypothesis test. The direct component result is much smaller: removing preference adaptation changes pooled hypervolume by 0.17%, and no preference-ablation contrast survives correction across the seven scenarios. Table 4 reports the full leaderboard, while the scenario-level and matched-budget analyses separate framework performance from preference-layer attribution. Comparisons with deterministic ranking rules remain descriptive and depend on the reported clipped full-front metric.

The four unresolved baseline contrasts all involve NSGA-II: benchmark_portfolio_optimization (Holm p = 0.064), budget_ranking_stability (p = 0.053), distribution_project_review (p = 1.0,

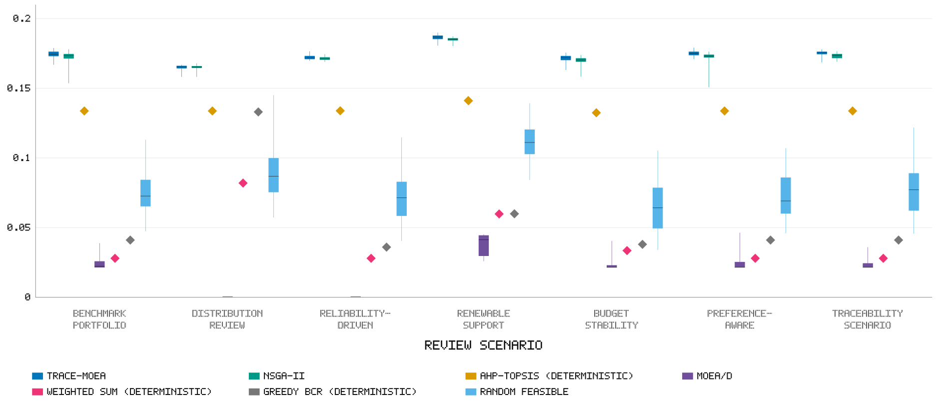

Figure 2. Hypervolume across the seven review scenarios (30 seeds per box for stochastic methods). AHP-TOPSIS, Weighted Sum, and Greedy BCR produce one portfolio per scenario and appear as diamonds. Random Feasible is stochastic. TRACE-MOEA and NSGA-II contend at the top of every scenario; the remaining baselines operate in a different performance regime.

Table 5. TRACE-MOEA versus NSGA-II per scenario (30 seeds). Confidence intervals are pointwise, multiplicity-unadjusted 5000-resample bootstrap intervals for the mean difference; r_{rb} is the rank-biserial effect. Holm-adjusted rank-test p-values determine significance across twelve stochastic opponents within each scenario.

Scenario	TRACE-MOEA	NSGA-II	Mean difference (95% CI)	r_{rb}	Holm p
benchmark_portfolio_optimization	0.17441	0.17224	0.00218 [0.00047, 0.00425]	0.376	0.0637
budget_ranking_stability (0.88 x B)	0.17139	0.16955	0.00184 [0.00026, 0.00353]	0.358	0.0529
distribution_project_review	0.16490	0.16512	-0.00022 [-0.00115, 0.00066]	-0.020	1.000
preference_aware_support	0.17512	0.17229	0.00283 [0.00138, 0.00477]	0.593	0.000407
reliability_driven_review	0.17223	0.17144	0.00080 [0.00010, 0.00153]	0.298	0.242
renewable_accommodation_review	0.18646	0.18505	0.00141 [0.00051, 0.00229]	0.504	0.00325
traceability_evaluation	0.17521	0.17325	0.00196 [0.00102, 0.00292]	0.560	0.00100

with a nominally negative difference of -0.13%), and reliability_driven_review (p = 0.242). Every comparison with the implemented R-NSGA-II configuration is significant. Figure 2 shows the overall distributions; Figure 5 isolates the direct preference-family control without compressing the closer TRACE-MOEA–NSGA-II differences.

Table 5 breaks down the TRACE-MOEA vs. NSGA-II comparison by scenario. The three Holm-significant wins occur exactly in the scenarios whose stakeholder weighting departs most from uniform (preference_aware_support, renewable_accommodation_review, traceability_evaluation). In the balanced full-pool scenarios (benchmark, budget stability) the margins remain positive (+1.1–1.3%) but do not survive Holm correction at n = 30. On the SimBench-only pool (distribution), the observed difference is nominally negative and not statistically significant; this is not an equivalence result.

The pattern is consistent with, but does not prove, the design hypothesis. Gains appear in scenarios with explicit preference emphasis; in balanced scenarios, differences are smaller and do not separate statistically.

6.2. Ablations: Where the Gain Does and Does Not Come From

Figure 3 orders the observed configuration contrasts. The largest differences occur for the `NoReliabilityFeatures` combined control (47.9% lower pooled hypervol-

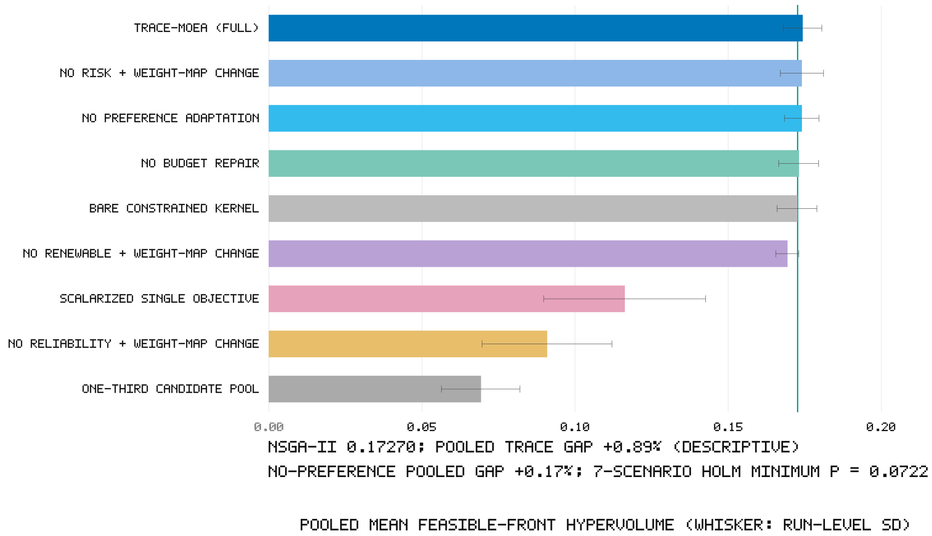

Figure 3. Pooled mean hypervolume of the full method and the eight ablations (error bars = one standard deviation). Two annotations identify the results that most constrain architectural interpretation.

ume; 0.09081 vs. 0.17425), the single-objective formulation (33.3% lower; 0.11617), the `NoRenewableFeatures` combined control (2.8% lower; 0.16930), and the one-third candidate pool (60.3% lower; 0.06920). The objective-hiding contrasts jointly change a search coordinate and the first-vector weight mapping, so they cannot identify the effect of objective visibility alone. The results show that these tested configurations materially affect the proxy task; they do not attribute the difference to a single named objective.

Among the component-level ablations, disabling budget repair (NoFeasibilityRepair) costs 0.72% of pooled hypervolume and reduces event–final-front candidate-position co-occurrence coverage from 0.986 to 0.865. This identifies the contribution of repair-drop positions to the set-overlap statistic; it does not validate human usefulness. Disabling all review-specific components at once (NSGA2Only, the bare kernel) costs 1.01% (0.17249 vs. 0.17425).

The component results constrain the algorithmic claim. Removing preference adaptation (`NoPreferenceRanking`) changes pooled hypervolume by only 0.17%. The full method wins the renewable scenario under the within-scenario family ($p_{Holm} = 0.0206$), but no preference-ablation contrast remains significant after a second Holm correction across the seven scenarios ($p = 0.0722$ for that cell). The `NoScheduleRisk` combined control changes pooled hypervolume by 0.13%. It is higher in `traceability_evaluation` ($p_{Holm} = 0.0219$ within scenario), but that contrast also falls just outside the cross-scenario threshold ($p = 0.0510$). Because the control changes both risk-objective visibility and the first-vector mapping, this is an exploratory configuration pattern rather than an isolated schedule-risk effect.

Two readings follow. The preserved schedule-risk control cannot distinguish a weak gradient from its altered preference-vector mapping, so no schedule-risk mechanism is assigned. Preference-selected best responses also overlap substantially with portfolios already preserved by constraint-dominated crowding, consistent with evidence that preference guidance is not automatically beneficial [20]. The complete TRACE-MOEA has the highest pooled hypervolume among the tested external baselines and no significant per-scenario loss, but its margin over its leanest viable variant is small. A hypervolume-only user could disable preference adaptation with little measured loss. The observed trade-off is a reduction in candidate-position co-occurrence coverage (0.966 versus 0.986). Because actual replacement and eviction are not counted, the release cannot show which weighting directions changed the population.

6.3. Implemented Run-Level Event Summaries and Candidate Co-Occurrence

Because no event statistic enters any ranking, the implemented event fields are reported descriptively. Across the 210 proposed-method runs, each released row reports event count and final-front candidate-position co-occurrence. TRACE-MOEA averages 1126.25 records per run (sample standard deviation 134.86): a mean 806.25 repair-drop records plus exactly 320 preference-best-response records (8 vectors $\times 40$ generations). The fixed 320 records are emitted whether or not population replacement is required; they are not an injection count. Mean co-occurrence is 98.6%, meaning that pool-local positions selected in returned-front portfolios usually also occur somewhere in the generated event-position set. This is a set-overlap statistic; it does not identify which event caused a portfolio or project position to survive.

The two event-bearing ablations bracket the sources. Without adaptive preference elitism (NoPreferenceRanking), the run records summarize only repair-drop positions and achieve mean coverage of 96.6%. Without the repair operator (NoFeasibilityRepair), each run retains the fixed 320 preference-best-response records and achieves coverage of 86.5%. The complete method's 98.6% is the joint descriptive set overlap of both event types. Because this is a combined ablation contrast, it supports only the joint event-type interpretation; it does not isolate the value of ordered history, which is absent from the release.

We do not claim that the event summaries improve review outcomes or that 98.6% is a desirable target. Validating whether records help review boards make faster or more consistent decisions would require a separately approved human study. Stable candidate identifiers, ordered payloads, replacement and eviction flags, and sufficient state snapshots would have to be serialized and tested before replay or project-level chronology could be claimed. Event variables are quarantined from evaluation; this does not imply zero computational or human-review cost.

Figure 4 makes event-record production measurable without turning it into an optimization objective. Removing preference elitism reduces both recorded event count and position co-occurrence. Removing repair eliminates repair-drop events and leaves the fixed preference-best-response records. The bare kernel has neither event channel. These are diagnostic differences in implemented records and set overlap; they do not imply that a larger record improves portfolio quality or human review.

6.4. Matched-Budget Preference Controls

The direct-control scan freezes the 120-candidate pool and balanced scenario weights, varies the budget multiplier over $\{0.75, 1.00, 1.25\}$, and runs TRACE-MOEA, R-NSGA-II, and NSGA-II for 30 independent seeds at each level. The normalized aspiration is fixed by those weights, while its raw objective-space point is recomputed from each budget's frozen bounds. The scan therefore adds 270 runs without reusing the heterogeneous scenario rows of the main experiment. Alongside hypervolume, it reports the positive-part Euclidean achievement distance from the entire feasible front to the declared normalized aspiration,

$$d_{\text{pref}}(\mathcal{P}) = \min_{x \in \mathcal{P}} \left\| \max\{\tilde{F}(x) - z^{\text{pref}}, 0\} \right\|_2,$$

where the maximum is componentwise and zero means that one front point meets every aspiration coordinate. Lower values indicate closer preference achievement. This

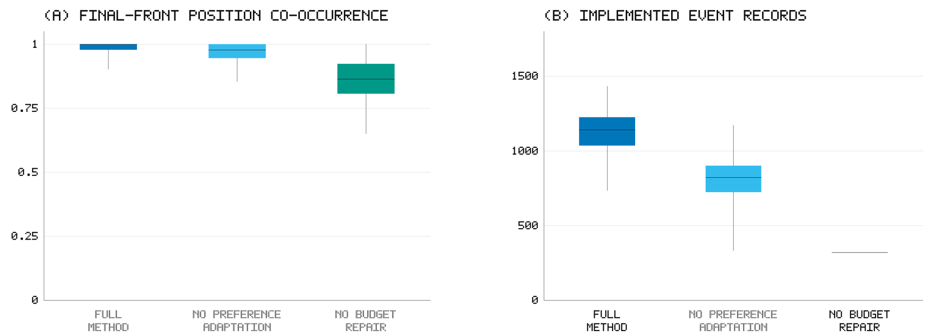

Figure 4. Implemented run-level event-record diagnostics pooled across seven scenarios and 30 seeds (210 runs per configuration). (a) Fraction of pool-local positions represented in the final feasible front that also occur in at least one generated event record; (b) number of records summarized in each run row. Boxes are descriptive. Event statistics are quarantined from all objectives, constraints, and selection decisions. The release does not establish lineage, chronology, or replay.

Table 6. Budget and direct preference-control scan (30 seeds per cell; lower preference distance is better).

Budget	Method	Mean HV	Mean preference distance	HV verdict versus TRACE
0.75 x B	TRACE-MOEA	0.16650	0.52985	–
0.75 x B	NSGA-II	0.16439	0.53252	TRACE win, Holm $p = 0.0327$
0.75 x B	R-NSGA-II	0.11007	0.55375	TRACE win, Holm $p < 10^{-9}$
1.00 x B	TRACE-MOEA	0.17509	0.52597	–
1.00 x B	NSGA-II	0.17304	0.53007	TRACE win, Holm $p = 0.00177$
1.00 x B	R-NSGA-II	0.12862	0.53431	TRACE win, Holm $p < 10^{-9}$
1.25 x B	TRACE-MOEA	0.17849	0.52323	–
1.25 x B	NSGA-II	0.17675	0.52764	TRACE win, Holm $p = 0.00144$
1.25 x B	R-NSGA-II	0.13563	0.52898	TRACE win, Holm $p < 10^{-9}$

second metric is descriptive and prevents a preference method from being judged only by global front coverage.

Hypervolume rises with budget for all three methods. Within each budget's two-comparator family, TRACE-MOEA significantly exceeds both controls after Holm correction. Its mean preference distance is also lowest at every level, but that ordering is descriptive because a second multiplicity family was not preregistered. R-NSGA-II's adverse result is retained as an implementation-specific direct control, not generalized to all reference-point algorithms.

6.5. Matched Output, Hypervolume Bounds, and Prespecified Sensitivity

The preserved matched-output analysis applies the same compromise rule already executed in the main pipeline and retains one selected portfolio per run. It therefore avoids comparing a many-point evolutionary front directly with a one-point deterministic output. Table 7 reports scenario-balanced means of the available compromise attributes. No row dominates the others: AHP-TOPSIS has the largest reliability total, Greedy BCR the largest renewable total, and MOEA/D the lowest risk and cost index largely because its compromise contains fewer than one project on average across scenarios. TRACE-MOEA occupies a different trade-off, not a universal optimum. The full-front hypervolume gaps

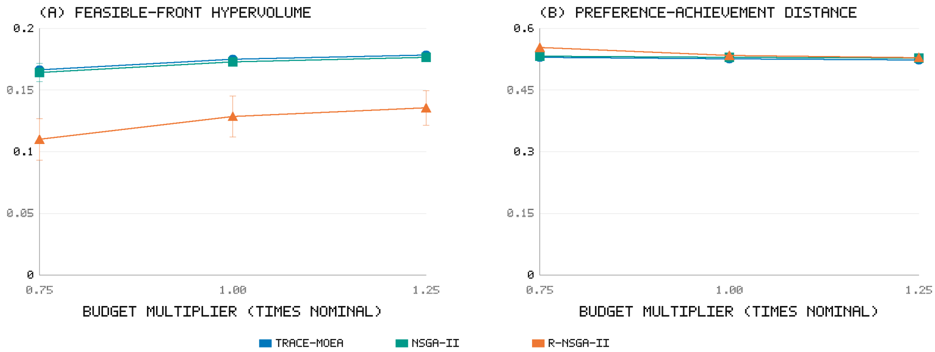

Figure 5. Hypervolume and preference-achievement distance for TRACE-MOEA, NSGA-II, and R-NSGA-II at 0.75, 1.00, and 1.25 times the nominal budget. Error bars show one standard deviation over 30 seeds. The two panels keep global front coverage and distance to the declared preference point conceptually separate.

Table 7. Matched one-output comparison using the preserved shared compromise rule. Stochastic rows average 30 seeds within each scenario and then seven scenario means; deterministic rows contain seven unique outputs. Objectives retain their native proxy scales.

Method	Type	Cost index	Reliability	Renewable	Risk	Portfolio size
TRACE-MOEA	stochastic MOEA	0.9136	25.5790	33.1729	0.2778	14.07
NSGA-II	stochastic MOEA	0.9341	24.7311	19.2924	0.2718	13.54
R-NSGA-II	stochastic MOEA	0.9779	14.8568	69.4005	0.2295	13.30
MOEA/D	stochastic MOEA	0.0581	0.3028	12.0026	0.1318	0.83
AHP-TOPSIS	deterministic	0.9866	35.1204	0.4155	0.3203	14.29
Weighted Sum	deterministic	0.9756	4.3834	41.7845	0.3664	5.43
Greedy BCR	deterministic	0.9715	7.6598	156.3509	0.2157	13.29

against deterministic rules remain valid for their reported metric but do not establish matched-output superiority.

The bound-sensitivity analysis identifies a load-bearing normalization choice. In the full-pool benchmark scenario, the padded empirical cost interval is $[-36881.68, 774515.23]$ even though the feasible budget is 1160, because the reference set includes unfiltered singletons. Table 8 reports the complete five-coordinate vectors for that scenario; all seven scenario vectors are included in the supplementary result tables.

Across the 16,216 non-dominated points returned by the 441 reruns, 1587 (9.79%) fall outside at least one reported empirical $[0, 1]$ coordinate and are clipped. The 25%-expanded bounds reduce this count to 154 (0.95%), while the analytic bounds have zero excursions beyond the 10^{-12} tolerance. Removing clipping changes values materially: Table 9 shows that AHP-TOPSIS exceeds TRACE-MOEA under the unclipped and expanded empirical schemes, whereas TRACE-MOEA exceeds it under analytic bounds. Thus the reported 29.4% full-front gap over AHP-TOPSIS is not normalization-invariant. TRACE-MOEA remains above NoPreferenceRanking under every tested scheme, but the difference ranges from 0.000291 under the reported clipped score to 0.000431 under analytic bounds at reference 1.1 and 0.001040 at reference 1.2. These are descriptive sensitivity differences; the existing corrected component test remains unresolved.

The one-factor-at-a-time scan further limits mechanism attribution. Relative to the registered TRACE cell (mean analytic-bound HV 0.105439 at reference 1.2), maximum rather than mean portfolio risk lowers mean HV by 6.88%, a 0.75 compliance share lowers it by 2.30%, $K = 16$ lowers it by 0.12%, and the reliability profile lowers it by 0.43%. The 0.25 compliance share, $K = 4$, renewable profile, and traceability profile increase the mean by

Table 8. Hypervolume bounds in the benchmark scenario. Coordinate order is cost, negative reliability, negative renewable support, risk, and negative quality.

Scheme	Lower-bound vector	Upper-bound vector
Reported empirical	$(-36881.678, -22.6339, -9308.767, 0.08$[illegible]	[illegible]$, 1.0778, 443.275, 1.04379, 0.045)$
25%-expanded empirical	$(-239730.906, -28.5618, -11746.777, -$[illegible]	[illegible]$, 2881.285, 1.28464, 0.2925)$
Conservative analytic	$(0, -124.4391, -11002.521, 0.1242, -0.901)$	$(1160, 0, 0, 1, 0)$

Table 9. Scenario-balanced mean hypervolume under bound, clipping, and reference-point sensitivity. The deterministic methods return one point per scenario.

Method	Reported clipped, ref 1.1	Reported unclipped, ref 1.1	Expanded unclipped, ref 1.1	Analytic, ref 1.1	Analytic, ref 1.2
TRACE-MOEA	0.174247	0.198818	0.213194	0.032181	0.118820
NoPreferenceRanking	0.173956	0.196180	0.211522	0.031750	0.117779
AHP-TOPSIS	0.134678	0.208164	0.226665	0.005168	0.028036
Weighted Sum	0.041048	0.041048	0.069712	0.001781	0.012253
Greedy BCR	0.055756	0.061968	0.091315	0.002664	0.016538

1.35%, 0.25%, 0.98%, and 0.41%, respectively. In the 0.75-compliance formulation, TRACE is 0.000795 below its formulation-matched NoPreferenceRanking control. The secondary one-output metric can move oppositely to full-front HV (for example, the renewable profile raises full-front HV but lowers matched single-point HV by 0.000727). Table 10 reports every prespecified cell; none was filtered by direction.

The new runs do not change the external-consistency evidentiary level. Neither NERC nor MTEP16 is rerun, and no portfolio-preserving randomization family is added; their associations therefore remain descriptive.

6.6. Public-Record Consistency Checks

Hypervolume measures optimization quality on the proxy objectives; it does not establish that selected portfolios identify real grid risk or predict project outcomes. We therefore add two external-consistency checks. Both are descriptive because the NERC rule overlaps the construction corpus and the MTEP project-level tests do not preserve portfolio dependence or a confirmatory comparison family.

Rung 1 — NERC rule consistency. Each candidate receives a priority score equal to a NERC-report-topic weight for its archetype multiplied by a stress percentile from raw RTS-GMLC and SimBench statistics. The score is computed before the pool's NERC-based attribute adjustment, preventing a direct mirror of that adjustment. Alignment is the mean priority of selected projects divided by the mean priority of the full pool; values above 1.0 indicate oversampling of documented-risk candidates.

TRACE-MOEA achieves priority capture of 1.55 in the benchmark scenario and 1.34 in the reliability-driven scenario. Its Kendall tau values are 0.01 and 0.15. AHP-TOPSIS posts the highest alignment (capture 2.29 and 1.63; tau 0.45 and 0.52). These are raw, unadjusted associations in a descriptive check. AHP-TOPSIS directly weights reliability-type attributes adjacent to the rule, so its high alignment partly reflects construct overlap. Greedy BCR provides an adverse descriptive control, with capture 0.23–0.53 and negative raw tau values.

Rung 2 — MISO MTEP16 outcomes. The second rung replaces a constructed rule with observable real-world outcomes. From the MTEP16 Appendix A and B project table (1218 projects), those with positive 2016 cost estimates form backtest pools of 1097 and 1062 projects for the two review scenarios. Features use only 2016-vintage fields (cost estimate, project type, voltage, mileage, appendix status, record date). Labels come from quarterly status snapshots (2016-12 through 2018-01) and the 2026 MISO in-service and active-project lists, yielding 924 built, 19 explicitly withdrawn, 39 deferred, and 236 unresolved projects

Table 10. Prespecified TRACE sensitivity relative to the registered cell; the last column compares TRACE with the matching NoPreferenceRanking formulation (or the registered control for preference-only changes). No p-values are computed.

Cell	Factor	Mean-HV change	Percent change	TRACE minus no-preference MOEA
Maximum risk aggregation	formulation	-0.007257	-6.88%	0.003155
Compliance share 0.25	formulation	0.001423	1.35%	0.000394
Compliance share 0.75	formulation	-0.002426	-2.30%	-0.000795
$K = 4$	preference	0.000262	0.25%	0.001475
$K = 16$	preference	-0.000128	-0.12%	0.001084
Reliability profile	preference	-0.000457	-0.43%	0.000755
Renewable profile	preference	0.001035	0.98%	0.002248
Traceability profile	preference	0.000433	0.41%	0.001646

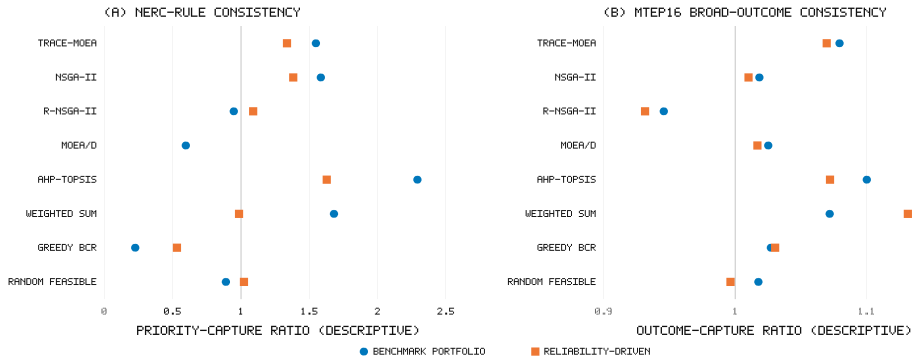

Figure 6. External-consistency ladder. (a) NERC rule backtest: priority-capture ratio against a rule combining NERC-topic archetype weights with NERC-independent physical stress percentiles (10 seeded compromise portfolios per method). (b) MISO MTEP16 outcome backtest: broad outcome-capture ratio against real built/withdrawn/unresolved project statuses. The vertical line marks parity with uniform random selection.

across the full table. No outcome field enters any feature, and no constant was fitted to outcomes.

TRACE-MOEA's broad capture is 1.079 in the benchmark scenario and 1.070 in the reliability-driven scenario. Raw project-level diagnostics are point-biserial $r = 0.169$ ($p < 10^{-7}$) and $r = 0.151$ ($p = 10^{-6}$), respectively; the corresponding unadjusted Mann–Whitney readouts point in the same direction. These p-values are descriptive diagnostics, not confirmatory tests: projects co-occur within selected portfolios, and no portfolio-preserving permutation family was preregistered. The capture ratios exceed those of the tested evolutionary baselines, but AHP-TOPSIS reaches 1.100 in the benchmark scenario and Weighted Sum reaches 1.132 in the reliability-driven scenario. Strict-label results use only 19 explicit withdrawals and are underpowered.

Five boundary conditions restrict the interpretation of Rung 2. First, the MTEP16 approved pool is approximately 98% built within the strictly labeled subset, so the backtest measures alignment within an already board-approved plan. Second, the `unresolved` class may contain projects re-scoped under new identifiers. Third, the pool contains almost no renewable or storage projects, so this rung provides no evidence for renewable-related objectives. Fourth, appendix status is decision-time information correlated with broad outcomes. Fifth, the raw project-level association tests do not preserve dependence induced by portfolio selection. Within these limits, the backtest shows descriptive contact with real

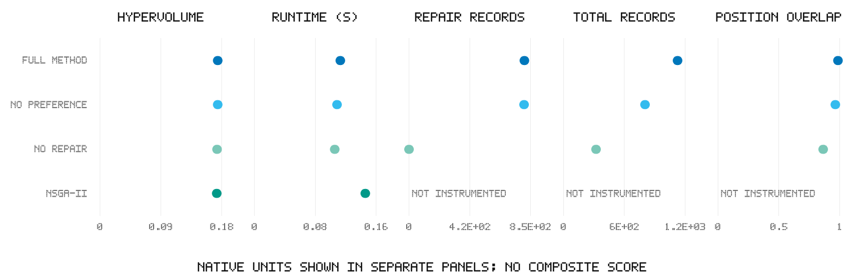

Figure 7. Descriptive pooled search-and-event profile over seven scenarios and 30 seeds per method. Hypervolume, runtime, repair-drop record count, total event-record count, and position co-occurrence retain their native units; the panels should not be read as a scalar composite score. NSGA-II's event fields are not instrumented.

outcomes; it does not establish above-chance portfolio performance, best alignment, or engineering-economic effectiveness.

6.7. Search-Effort and Outcome-Sensitivity Diagnostics

Figure 7 relates optimization quality to computation and implemented event-record production. TRACE-MOEA averages 0.17425 hypervolume, 0.1130 s, 806 repair-drop records, 1126 total generated records, and 0.986 position co-occurrence per run. Removing preference elitism leaves 0.17396 hypervolume and about 803 repair-drop records; disabling repair leaves 320 preference-best-response records and lowers co-occurrence to 0.865. NSGA-II attains 0.17270 at 0.1456 s without an instrumented event list. Event-record count remains descriptive and is not an optimization objective.

The diagnostic clarifies the practical trade-off. TRACE-MOEA is faster than the tested NSGA-II implementation at this small problem size, but deterministic baselines remain orders of magnitude cheaper. Its additional released fields are a run-level record count and candidate-position set overlap. The 98.6% value means that almost every pool-local position represented in the returned fronts also occurs somewhere in the generated event-position set. It is not a released chronology, stable-identifier lineage, replay evidence, or evidence that human reviewers find the summaries sufficient.

Figure 8 re-expresses the MTEP16 result without collapsing broad and strict labels. Broad capture is 1.079 and 1.070 in the benchmark and reliability-driven scenarios, respectively, whereas strict capture is 1.014 and 1.000. The broad view supplies statistical power by treating unresolved projects as non-built; the strict view avoids that uncertain label but contains only 19 withdrawals in the full inventory. Their divergence is not a robustness success: it quantifies the outcome-definition sensitivity that bounds the external-validity claim.

The two diagnostics answer different questions. Figure 7 shows the computation and run-level event volume associated with the proxy-objective result, whereas Figure 8 shows how the external-consistency result depends on the definition of the negative class. Their aggregate tables are retained under `manuscript/derived_tables/` for independent recomputation.

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7. Discussion

Framework performance does not identify a preference-layer benefit. TRACE-MOEA is descriptively above NSGA-II in the pooled proxy summary, and the significant scenario-level differences occur in preference-emphasized settings. The direct `NoPreferenceRanking` ablation, however, remains unresolved after correction across seven

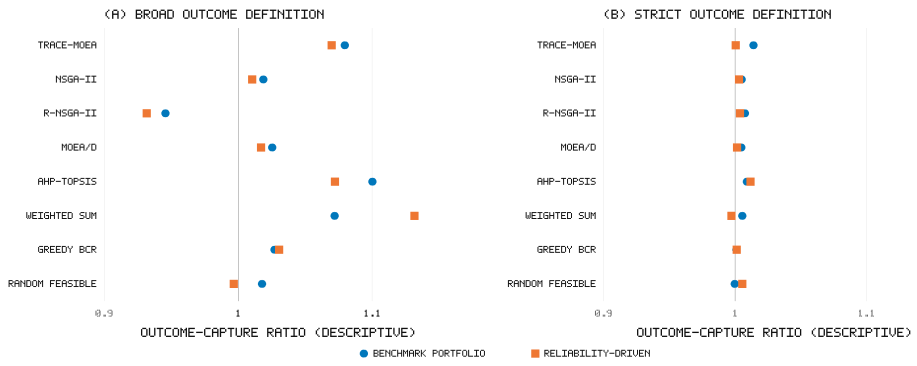

Figure 8. MTEP16 outcome-capture ratios for the two review scenarios. The dashed line denotes parity with uniform selection. Broad labels include unresolved projects among negatives; strict labels compare built projects only with explicit withdrawals. Proposed-method highlighting is visual identification and does not override the reported significance tests.

scenarios. Preference-selected best responses may overlap portfolios already preserved by constraint-dominated crowding, which is consistent with the small marginal difference but is not a proved mechanism. The prespecified scan contains favorable, adverse, and near-null cells, so update cadence, vector count, and seed profiles remain design variables rather than validated sources of gain.

Normalization and output matching. The reported clipped empirical metric mixes feasible random portfolios with unfiltered singleton bounds. Clipping affects 9.79% of rerun front points, and the deterministic-rule ordering changes under expanded versus analytic bounds. The proposed method remains slightly above NoPreferenceRanking in the tested schemes, but that stability does not transfer to a general claim against point-valued MCDA rules. The matched one-output comparison also shows distinct reliability, renewable, risk, cost, and cardinality trade-offs rather than a dominating recommendation. Hypervolume is appropriate for front coverage; it should not be read as a matched decision-output utility score.

The event summaries support inspection at the run level only. TRACE-MOEA supplies event count and final-front position-co-occurrence fields unavailable from the comparison methods. These summaries show which pool positions occur in both event and returned-front sets, but the release preserves neither stable identifiers, order, replacement flags, evictions, nor population snapshots. It therefore cannot reconstruct a project-level intervention sequence. A future lineage claim requires serialized state transitions, while a decision-support claim requires human evaluation.

Public-record consistency provides a secondary check. AHP-TOPSIS is favored on the NERC view partly because its reliability attributes overlap the priority rule. The MTEP view measures capture within an already-approved plan and is constrained by its high build rate. The diagnostics show that optimized portfolios retain some alignment with public records without establishing review accuracy.

The direct preference-family control narrows the comparison. TRACE-MOEA is higher than the implemented R-NSGA-II configuration in both front coverage and mean preference distance across the three matched budgets. R-NSGA-II uses one declared reference point and one matched configuration, so the result does not generalize to all reference-point methods. Combined with the unresolved direct ablation, it supports the complete framework under the disclosed setup without isolating adaptive preference elitism as the cause.

For a planning workflow, TRACE-MOEA offers budget-feasible fronts and run-level diagnostics at modest computational cost. The matched one-output attributes and normalization checks do not identify a universal winner. Its practical role is therefore constrained portfolio exploration with inspectable summaries, pending expert calibration, electrical validation, and a human study of review value.

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8. Limitations

Six limitation groups constrain the scope of the claims in this paper.

1. **The proxy task lacks expert review ground truth.** The main benchmark evaluates algorithmic performance on engineering proxies rather than expert-labeled review outcomes. NERC and MTEP16 provide descriptive consistency checks, but the former shares kind-level constructs with pool generation and the latter evaluates projects in an already approved plan with approximately 98% strict-label build rate and only 19 explicit negatives. An independently annotated subset with inter-rater agreement is needed to evaluate review correctness.
2. **Economic, electrical, and benchmark validity remains limited.** Candidate costs and benefits are synthetic functions of network aggregates, all seven scenarios use one 120-candidate pool, and selected portfolios have not been checked by AC power flow. A second independently constructed pool, calibration against published utility investment data, and OPF or load-flow evaluation are required before economic or electrical recommendations.
3. **Preference-layer attribution remains incomplete.** Removing preference adaptation changes pooled hypervolume by only 0.17% and the corrected direct effect is unresolved. The sensitivity scan varies K and four seed profiles on one scenario but not update cadence, perturbation scale, eviction, penalty, or pool size. R-NSGA-II supplies one matched preference-based comparator, not a survey of RVEA, preference-aware NSGA-III, or interactive methods. The evidence therefore supports the complete framework against the implemented controls without isolating a general preference-adaptation benefit.
4. **Metric and ablation interpretation depends on implementation choices.** The empirical hypervolume reference includes unfiltered singleton portfolios, its cost upper bound is far above the feasible budget, and 9.79% of rerun front points require clipping. Alternative recomputations preserve the small TRACE–NoPreference order but change deterministic-rule ordering. In addition, objective-hiding controls also alter preference-vector mapping, so they estimate combined perturbations rather than isolated objective contributions.
5. **Run-level event summaries do not establish lineage or review utility.** Event statistics are quarantined from optimization, and the released 98.6% co-occurrence is a set-overlap measure over pool-local positions. Stable identifiers, event order, replacement or eviction flags, and population snapshots are absent. Project-level lineage requires serialized state transitions, while claims about review speed, contestability, or decision quality require human evaluation.
6. **Exact baseline-state replay is not available.** The original R-NSGA-II record preserves the reference point, `epsilon=0.01`, population, generations, and returned rows, but not the pymoo version, all Boolean-operator defaults, internal normalization state, or per-generation ideal and nadir values. A compatible rerun could not be certified, so the matched-output analysis uses the preserved comparator rows and applies only to that disclosed implementation.

9. Conclusions

TRACE-MOEA integrates constrained portfolio search, preference-guided elitism, deterministic budget repair, and quarantined run-level event summaries. On the five-objective, 120-project proxy benchmark, the complete framework attains descriptive pooled mean hypervolume 0.17425, 0.89% above NSGA-II. The matched-budget preference comparison also favors TRACE-MOEA over the implemented R-NSGA-II configuration, while the one-output analysis shows competing objective trade-offs rather than a universally preferred recommendation.

The component evidence is narrower than the framework comparison. Removing preference adaptation changes pooled hypervolume by 0.17%, and its direct effect remains unresolved after correction across seven scenarios. Normalization changes the deterministic-rule ordering, so the reported full-front metric does not support a normalization-invariant cross-family conclusion. The NERC and MTEP16 analyses provide descriptive external consistency rather than validated review accuracy.

Run-level event count and final-front position co-occurrence make event production measurable without changing optimization. The 98.6% co-occurrence value is a set-overlap summary, not evidence of lineage, replay, or human-review value. TRACE-MOEA should therefore be interpreted as an integrated budget-constrained proxy-search framework with measurable run summaries. Stable identifiers, ordered state transitions, expert labels, electrical checks, and human evaluation are required before stronger claims about chronology, review effectiveness, or deployment.

Author Contributions

[AUTHOR INPUT REQUIRED: assign the CRediT roles to Yubin Lin, Jiyu Li, Xiaofei Ruan, Xiaoyu Huang, and Dishan Yang, and obtain approval from every author.] All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

All data sources used in this study are publicly accessible. RTS-GMLC source data is available at https://github.com/GridMod/RTS-GMLC. The SimBench complete mixed dataset is available at https://simbench.de. NERC reliability reports are available at https://www.nerc.com; only report metadata is used in this study, and a manifest with official URLs and SHA-256 checksums is released in place of redistributed PDF files. MISO MTEP16 Appendix A and B project records, subsequent quarterly status snapshots, and the 2026 MISO in-service and active-project portal lists are available at https://www.misoenergy.org.

The candidate-derivation pipeline, TRACE-MOEA implementation, baseline and ablation configurations, 3360 main per-run records, the 270-run three-budget control scan, inference tables, public-record analyses, figure scripts, and the stage-local matched-output/normalization/sensitivity package are included in the supplementary package and are available from the corresponding author. The package contains 861 matched-output analysis rows, 441 front-level bound/reference reruns, 390 sensitivity runs, and an independent same-seed reproduction. The original machine-readable configuration records population, generations, methods, and evaluation but is not a complete hyperparameter manifest. Main run rows contain only event-record count and final-front pool-position co-occurrence, not payloads, stable IDs, replacement flags, or state history. A persistent public repository can be supplied before publication, subject to third-party redistribution terms. The companion project `mintou_p6_bilonsga_project_review` (BiLo-NSGA [28]) shares candidate generation, source corpora, common benchmark and evaluation utilities, and public-record analysis infrastructure. Paper-specific configurations, executions, run outputs, selected portfolios, comparisons, and claims are not shared evidence.

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During the preparation of this manuscript, the authors used generative AI tools (Claude, Anthropic) for language refinement, literature summary, and formatting assistance, under the authors' full intellectual oversight. The authors have reviewed and edited all AI-assisted content and take full responsibility for the final manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

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